

Solving Linear Equations with Variables on Both Sides

Reading a balance or table model as an equation, collecting the variable terms on one side, and distributing before you solve

How to work each problem.

- **The balance model.** In the pan pictures, $\square$ is a sealed bag holding the same unknown number of counters — call it x — and $\bullet$ is one loose counter. The pans balance, so the two sides are equal. Removing the *same* thing from both pans keeps the balance; that is exactly what a legal move on an equation does.
- **Collect, then isolate.** Move the variable terms to one side and the constants to the other by adding or subtracting the same quantity on both sides. Then divide by the coefficient.
- **Distribute first** when a side is a product, such as $2(x + 5)$.
- **Check every answer** by substituting it back into the *original* equation and evaluating both sides.
- Show the balanced move beside each line, for example “ $- 2x$ on both sides”.

1. Read the balance, then solve. The pans below balance.

Left pan		Right pan
$\square \square \square \bullet \bullet$	$=$	$\square \bullet \bullet \bullet \bullet \bullet \bullet \bullet \bullet$

Write the equation the picture shows, then take one bag off each pan and finish solving for x . Equation: $3x + 2 = x + 8$.

Answer: _____

2. Solve for x , showing the move you make on each line.

$$5x - 4 = 2x + 11$$

Answer: _____

3. Read the balance, then solve. The pans below balance.

Left pan		Right pan
$\square \square \square \square \bullet$	$=$	$\square \square \bullet \bullet \bullet \bullet \bullet \bullet \bullet \bullet \bullet \bullet$

Write the equation the picture shows, then remove two bags from each pan and finish solving.
Equation: $4x + 1 = 2x + 9$.

Answer: _____

4. Distribute first, then solve for x .

$$2(x + 5) = 4x - 6$$

Answer: _____

5. Solve for x . Watch the signs when you move a negative term.

$$8x + 1 = 3x - 14$$

Answer: _____

- 6.** Solve for x . The answer is not a whole number — leave it as a fraction in lowest terms.

$$3(2x - 1) = 4x + 8$$

Answer: _____

- 7.** Solve for x , and write the answer as a fraction in lowest terms. Then check it by substituting into both sides of the original equation.

$$6x - 5 = 2x + 4$$

Answer: _____

8. Two plans, one break-even point. A club is choosing between two ways to pay for a season of practice sessions. Let x be the number of sessions.

Plan	Total cost, in dollars
Plan A: pay a joining fee, then a low rate per session	$40 + 12x$
Plan B: no joining fee, higher rate per session	$20x$

Write the equation that says the two plans cost the same, then solve it. How many sessions is that?

Answer: _____

9. What happens here? Distribute the left side, then collect the variable terms.

$$3(2x + 5) = 6x + 15$$

Write the equation you are left with after collecting, and describe every number that makes the original equation true.

Answer: _____

10. Challenge — and justify it. Distribute the left side, then collect the variable terms.

$$4(x + 2) = 4x + 5$$

Write the equation you are left with after collecting. Describe the solution set, and explain in one sentence why the variable disappearing tells you what it does.

Answer: _____